# Sequence-to-Expression Models Fail to Generalise to New Genomic Neighbourhoods

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## Abstract

Machine learning for synthetic biology is moving beyond interpreting DNA towards designing it. This shift requires verifiers, i.e. models that predict how designed DNA sequences will be expressed before they are constructed, but it remains unclear whether current sequence-to-expression (S2E) models can perform this role outside native genomic contexts. Baker's yeast provides a uniquely powerful eukaryotic testbed for this question: its genetic tractability enables systematic promoter swaps, genome-wide integrations, and large structural rearrangements that remain difficult to construct at comparable scale in mammalian genomes. Here, we benchmark two S2E models zero-shot across eleven yeast tasks. Both models approximately recover the experimental ordering of short, gene-proximal variants, but perform at approximately chance level on genome-wide reporter integrations and poorly on structural rearrangements. For native-promoter swaps, their predicted transcript-abundance ranges are only approximately two-fold, whereas measured fluorescence spans 66-fold and 481-fold for two reporter genes. These results identify contextual generalisation and dynamic-range compression as major obstacles to using current S2E models as verifiers for genome design. An open-source implementation of our benchmark can be found online.[1]

## 1 Introduction

The next frontier for machine learning in genomics is in designing DNA and not merely interpreting it. Models intended to guide genome engineering must therefore predict how genes behave under DNA sequence perturbations beyond the native genomic contexts that make up their training data. Baker's yeast offers a powerful eukaryotic testbed for this problem because its genome is far easier to engineer systematically and at scale than the human genome.

Yeast genetics established foundational principles of centromere function, DNA replication, and cell-cycle control [1, 2, 3], and its experimental advantage now extends to synthetic genomics. Sc2.0, the synthetic yeast genome project, enables deletions, duplications, inversions, and translocations to be generated throughout synthetic yeast chromosomes. Using these rearrangements, *Brooks et al.* [4] showed that transcriptional units are not context-independent parts: neighbouring transcription alters expression levels and transcript isoform boundaries, and convergent transcription can tune 3′-UTR length. Yeast therefore makes it possible to test whether sequence-to-expression models learn regulatory principles that remain valid when genomes are engineered rather than merely observed.

Genomic sequence-to-expression (S2E) models predict RNA-seq and related expression readouts directly from DNA sequence. If reliable under cis-perturbations, they could become core tools for synthetic biology: tuning the expression of engineered pathways and serving as predictive oracles that screen or steer sequences generated by DNA design models before they are constructed [5].

[1] https://github.com/anon-schmoo/neurips-icbinb-2026

Yet, in the native genomic data that dominate current training sets, most genes occur in only one neighbourhood. Large receptive fields make the surrounding sequence available to a model without forcing it to learn its causal contribution.

Here, we exploit yeast's genetic tractability to benchmark *Shorkie* [6] and *Yorzoi* [7] zero-shot across eleven tasks, ranging from single-nucleotide variants and MPRAs to promoter swaps, genome-wide reporter relocations, structural rearrangements, and exogenous bacterial chromosomes. We find that models rank short, gene-proximal variants, particularly in promoters and terminators, correctly, but predict severely compressed expression ranges (promoter swaps). Both perform at approximately chance level when identical reporter cassettes are integrated at new genomic sites and remain weak on structural rearrangements. We show that predictive features of position effects, such as reporter distance to Micro-C determined boundaries, are captured internally by *Yorzoi* but are nevertheless not converted to accurate expression predictions. Together, these results identify generalisation to distant sequence neighborhoods and quantitative response as central obstacles to using current S2E models for genome design.

## 2 Related Work

**S2E model benchmarks** *Karollus et al.* [8] benchmark mammalian S2E models directly across a variety of tasks, mainly population studies and deep perturbation assays. Their key finding is that mammalian S2E models at the time, such as *Enformer* [9], were substantially more accurate at predicting the effects of promoter perturbations but systematically underestimated the effects of distal enhancers. In contrast, we leverage the greater variety of genetic perturbations possible in yeast and benchmark more recent models that explore novel training methods or datasets.

**Yeast S2E models for RNA-seq** *Chao et al.* [6] and *Schneider et al.* [7] introduce some of the benchmark tasks presented here to evaluate the generalisation of their models, *Shorkie* and *Yorzoi*, to downstream tasks, including eQTL studies and promoter massively parallel reporter assays. However, with the exception of one task, they do not benchmark their models on the same tasks and do not cover the variety of genetic perturbations found in the yeast literature, such as perturbations of other regulatory regions, engineered structural variants, or genome-wide reporter translocations. Our benchmark fills this gap.

**Assay-specific S2E models** For many of the benchmark tasks we present, the authors trained small convolutional, decision-tree-based, or linear models to predict the outcomes of their reporter assays. For example, *Hong et al.* [10] train a small XGBoost model to predict the fluorescence of a reporter gene integrated at different genomic locations. *Rafi et al.* [11] evaluate a variety of deep-learning models, some contributed by the wider community, to predict the outcome of their massively parallel promoter assay. Although sometimes useful for uncovering explanatory features, these models are usually constrained to the narrow experimental datasets on which they were trained and are generally not applicable to other assays. Here, we benchmark S2E models trained on RNA-seq and additional sequencing-based assays that, in theory, should be able to generalise to other assays.

## 3 A benchmark for genomic sequence-to-expression models in baker's yeast

We introduce eleven benchmark tasks that probe different capabilities of S2E models in a zero-shot setting.

**S2E models** map a DNA sequence $x \in \{A, C, G, T\}^L$ to non-negative coverage predictions

$$\hat{y} = f(x) \in \mathbb{R}_{\geq 0}^{T \times C}, \tag{1}$$

where $f$ is the model, $T$ is the number of tracks, and $C$ is the number of predicted positions. Rows correspond to tracks, and columns correspond to positions. We primarily evaluate RNA-seq tracks: *Shorkie* [6] predicts one unstranded track per experiment, whereas *Yorzoi* [7] predicts separate forward- and reverse-strand tracks. Typically, $C < L$ because predictions are restricted to the central region of the input, reducing boundary effects from unobserved flanking sequences.

**Zero-shot evaluation** For each assayed construct, we reconstruct the tested sequence $x$, predict both $x$ and its reverse complement, invert the model-specific output transformation, reorient the reverse-complement prediction to $x$, and average the two predictions in coverage space. We then average all RNA-seq tracks for *Shorkie* or only those matching the reporter strand for *Yorzoi*, yielding one coverage profile $\tilde{y}$. For a forward-oriented reporter, this means using only *Yorzoi*'s forward-strand tracks. When a scalar proxy for reporter transcript abundance is required, we sum over its coding sequence:

$$y_r = \sum_{j=s}^{e} \tilde{y}_j, \tag{2}$$

where $s$ and $e$ are the coding-sequence boundaries.

### 3.1 S2E models fail to predict gene expression of genome-wide integrated reporter genes

**Motivation** The core result from [12], [13], and [10] is that the same gene expression cassette, complete with a promoter, coding sequence, and terminator, is differentially expressed when integrated at different positions in the genome. Hong et al. [10] found that the distance from the integration site to the nearest boundary of a self-associating chromatin domain [14] explains part of this variation. Although the models are not trained directly on Micro-C data to predict such boundaries, gene-expression neighbourhood effects could be learned from RNA-seq alone, which lies squarely within the existing models' training data.

**Task** We task the models with predicting the expression of a gene expression cassette containing a promoter, reporter CDS, and terminator at two types of integration site. Expression is experimentally measured as fluorescence. We correlate the predicted transcript abundance of the reporter with the experimental fluorescence measurement.

**Data** *Hong et al.* [10] integrate a red fluorescent protein expression cassette at 130 intergenic regions (one integration per genome) and measure fluorescence by flow cytometry. In contrast, *Wu et al.* [13] choose a different type of integration site and insert their reporter cassette into different strains of the yeast single-knockout collection at the positions of previously deleted non-essential genes. We note that some of the variation in this assay might be better explained by variation in the trans-regulatory environment (e.g., changes in transcription-factor abundance) than by the local position effect.

**Metrics** For each insertion site, we reconstruct sequence $x_i$ as the concatenation of the *upstream neighbourhood at i*, *fixed reporter cassette*, and *downstream neighbourhood at i*. We predict expression for each sequence:

$$X = \left\{ \text{concat}\left( \text{genome}_{i-\frac{L-R}{2}:i}, \text{reporter}, \text{genome}_{i:i+\frac{L-R}{2}} \right) \mid i \in I \right\} \tag{3}$$

$$\hat{Y} = \left\{ f_{i:j}(x) \mid x \in X \right\} \tag{4}$$

where $I$ is the set of surveyed integration sites, $\text{genome}_{m:n}$ is the genomic sequence from position $m$ to $n$, $R$ is the length of the reporter sequence, $L$ is the model input length, and $i$ and $j$ are the start and end positions of the reporter coding sequence, respectively. Performance is then reported as Pearson $r$ and Spearman $\rho$ for $\left\{ (\hat{y}_i, y_i) \mid y_i \in Y, \hat{y}_i \in \hat{Y} \right\}$, where $Y$ is the set of true fluorescence measurements of the reporter at each insertion site.

**Results** In both assays, Yorzoi and Shorkie have approximately random performance and fail to predict the effect of inserting the reporter cassette at a new site (see Table 1). *Hong et al.* [10] fit an XGBoost model to part of their data and determine that, among their hand-crafted features, the distance from the insertion site to the nearest Micro-C boundary is the most predictive (single-feature $r^2 \approx 0.1$). Here, a Micro-C boundary denotes a local position with the minimum number of crossing Micro-C interactions considered within a 10 kbp window (see Section A.5). We wondered to what extent the evaluated models capture this feature. It is important to note that both Shorkie and Yorzoi broadly follow a U-Net architecture in which the sequence is first downsampled to a shorter length, processed through transformer blocks with bidirectional self-attention, and then upsampled to sequence length while the number of channels is expanded to produce an embedding. Finally, a linear head transforms this embedding to match the target number of channels. To investigate the

Table 1: Zero-shot performance on the chromosomal position-effect benchmarks. Wu evaluates 1,043 scorable ORF-replacement loci. Hong reports the held-out IntProp set as the primary test set ($n = 52$) and IntTrain as a secondary split ($n = 98$). All correlations are close to zero or negative, indicating that neither model captures genomic position effects.

| Assay | $n$ | Metric | Shorkie | Yorzoi |
|---|---|---|---|---|
| Wu et al. | 1,043 | Pearson $r$ | –0.039 | 0.012 |
| | | Spearman $\rho$ | –0.029 | 0.016 |
| Hong et al. | 52 | IntProp Spearman $\rho$ | –0.148 | 0.063 |
| Hong et al. | 98 | IntTrain Spearman $\rho$ | –0.269 | –0.073 |

presence of the aforementioned feature, we trained linear probes on outputs of *Yorzoi* at three layers: the output of the downsampling convolutions, the output of the bottleneck transformer blocks, and the output of the upsampling convolutions before the output head [6, 7]. The target was the distance to the nearest boundary at each position in the sequence. Interestingly, the downsampling layers had no linearly decodable notion of the boundary distances, with a low Pearson correlation of $r = 0.16$ on held-out sequences. The output of the transformer layers, however, achieved a higher correlation of 0.46 on held-out sequences, demonstrating that some version of this feature is learned during training. In a later layer, this feature is apparently diluted, as the linear probe trained on the output of the upsampling layers achieved a lower Pearson correlation of $r = 0.31$. Our hypothesis is that this feature is not essential for predicting training windows and is thus only weakly learned during training. In follow-up experiments, we intend to retrain Yorzoi with Micro-C boundary distances or contact maps as an additional output, which would force the model to retain the feature and potentially improve performance on integration assays such as that of *Hong et al.*.

### 3.2 S2E models poorly predict expression changes after structural rearrangements

**Motivation** Although the native yeast genome offers limited diversity for testing how the expression of the same coding sequence changes across structurally distinct genetic neighbourhoods, engineered strains from the Sc2.0 project provide extensive structural variation. In contrast to other benchmark tasks that use a fluorescent marker, an *in-distribution* readout allows model failures to be attributed more clearly to the model rather than to inherent differences between the biological mechanisms captured by the assay and the model.

**Task** Models are benchmarked on their ability to predict the log-fold change in expression, quantified by long-read direct RNA-seq, for a gene in two different genetic neighbourhoods: the reference sequence REF on an unrecombined chromosome 9 (right arm) and the same gene in a recombined and therefore structurally rearranged genetic neighbourhood (ALT).

**Data** *Brooks et al.* [4] profile transcription in approximately 70 yeast strains with a structurally rearranged chromosome 9 (right arm) using long-read Nanopore RNA-seq. These rearrangements are produced by inserting recombination sites between the stop codon and 3′ UTR and then inducing recombination. This produces various structural variants, including deletions, duplications, inversions, and translocations. Each structural variant necessarily changes the 3′ UTR of the gene of interest. This dataset yields 215 unique sequences from 30 genes across approximately 50 strains, totalling 327 samples.

**Metric** We compute the true and predicted log-fold changes (LFCs) in gene expression and report balanced accuracy, Spearman $\rho$, and Pearson $r$. For many samples, we obtain more than one reference gene expression measurement from repeated experiments. In such cases, we first compute the mean LFC and then compute performance metrics from these means. Formally,

$$\mathrm{LFC}_i = \frac{1}{|Y_{\mathrm{ref},i}|} \sum_{y \in Y_{\mathrm{ref},i}} \log_2 \left( \frac{y_{\mathrm{alt},i}}{y} \right) \tag{5}$$

for gene $i$. Predicted LFCs are calculated by predicting coverage for the ref and alt sequences and summing over the coding-sequence bins.

Table 2: Results for *Brooks et al.* [4], log-fold-change (LFC) prediction on the shared cohort ($n = 327$). Bold denotes the best-performing sequence-to-expression model.

| Metric | Shorkie | Yorzoi |
|---|---|---|
| LFC direction balanced accuracy | 0.53 | **0.64** |
| LFC Spearman $\rho$ | 0.04 | **0.39** |
| LFC Pearson $r$ | –0.07 | **0.32** |

**Results** Neither Shorkie nor Yorzoi can accurately predict the effect of structural variation on gene expression. Although Yorzoi outperforms Shorkie, its balanced accuracy in predicting the correct direction of the effect is only slightly above 60%. Unlike Shorkie, Yorzoi was trained on data from [4], although the benchmark sequences scored here were held out from both models. These results generally point to the failure of current models to generalise to new sequence neighbourhoods that influence expression. Some of this performance difference could be attributable to the different RNA-seq assays used to train Shorkie and Yorzoi: while the benchmark uses long-read Nanopore RNA-seq, Shorkie is trained only on short-read Illumina RNA-seq.

### 3.3 S2E model predictions weakly correlate with true coverage of exogenous bacterial DNA

**Motivation** None of the other benchmark tasks tests the models on the task for which they were trained: RNA-seq track prediction. Testing on the native genome is difficult because Shorkie and Yorzoi have different, potentially overlapping training and test sets. *Meneu et al.* [15] transform two bacterial chromosomes into yeast and profile their expression using RNA-seq, providing a clean held-out test set comprising 2 Mbp of DNA sequence. Because one of the bacterial chromosomes has a substantially lower GC content, this experiment further tests the models' ability to generalise to out-of-distribution sequences.

**Task** Both models predict RNA-seq coverage profiles in a zero-shot setting for the two bacterial chromosomes, which are then compared with the measured RNA-seq coverage profiles.

**Data** *Meneu et al.* build two yeast strains, each with one additional linearised chromosome. One chromosome comes from *M. pneumoniae*, is roughly 818 kbp long, has a yeast-matched GC content of 40%, and is largely transcribed. The other, longer chromosome comes from the bacterium *M. mycoides*, is approximately 1.2 Mbp long, and has a lower GC content of 24%, which also substantially decreases its average expression.

**Metrics** Because the input sequence window of neither Shorkie nor Yorzoi spans the full genomic sequence, we tile each bacterial genome according to the output-window length of the respective model (Yorzoi: 3 kbp; Shorkie: approximately 14 kbp) and predict coverage profiles using the full input window (i.e., the sequence corresponding to the respective tile plus flanking sequences extending to the boundaries of the input window). Each model's predictions are compared with the ground truth using two metrics: bin-wise Pearson correlation and Jensen–Shannon divergence of the normalised profiles. Here, normalising a profile means dividing each predicted value by the sum of the bins in a given window so that the bins sum to 1. Formally,

$$\boldsymbol{y} \in \mathbb{R}^{L}_{\geq 0}, \quad \hat{\boldsymbol{y}} := f(x) \in \mathbb{R}^{L}_{\geq 0}, \tag{6}$$

where $x$ is the input sequence and $f$ is the respective model. Furthermore,

$$\boldsymbol{p} := \frac{\boldsymbol{y}}{\|\boldsymbol{y}\|_1}, \quad \hat{\boldsymbol{p}} := \frac{\hat{\boldsymbol{y}}}{\|\hat{\boldsymbol{y}}\|_1}, \quad \boldsymbol{p}, \hat{\boldsymbol{p}} \in \Delta^{L-1}. \tag{7}$$

where $\boldsymbol{y} \neq \mathbf{0}$, $\hat{\boldsymbol{y}} \neq \mathbf{0}$, and $\Delta^{L-1}$ denotes the $(L-1)$-dimensional probability simplex.

$$r_{\text{bin}} := r(\boldsymbol{y}, \hat{\boldsymbol{y}}), \quad D_{\text{JS}} := \text{JSD}(\boldsymbol{p}, \hat{\boldsymbol{p}}). \tag{8}$$

where $r$ is the usual bin-wise Pearson correlation.

**Results** Yorzoi consistently outperforms Shorkie across both bacterial chromosomes and all evaluated dimensions (see Table 3). Its median per-window Pearson correlation is higher and its Jensen–

Table 3: Results for *M. mycoides* (Mmmyco) and *M. pneumoniae* (Mpneumo). Bold denotes the best-performing model. JSD refers to Jensen–Shannon divergence; ↓ indicates that lower values are better.

| Benchmark | Metric | Shorkie | Yorzoi |
|---|---|---|---|
| ***M. mycoides*** **(Mmmyco)** | | | |
| Overall | Bin-wise Pearson | 0.2638 | **0.3843** |
| Overall | Shape JSD ↓ | 0.2844 | **0.2382** |
| ***M. pneumoniae*** **(Mpneumo)** | | | |
| Overall | Bin-wise Pearson | 0.0912 | **0.1103** |
| Overall | Shape JSD ↓ | 0.0982 | **0.0826** |

Shannon divergence is lower in every comparison. Nevertheless, the absolute correlations remain low: even Yorzoi reaches only $r_{\text{bin}} = 0.38$ on *M. mycoides* and $r_{\text{bin}} = 0.11$ on *M. pneumoniae*. *M. mycoides* has approximately three-fold higher Pearson correlation than *M. pneumoniae* for both models, despite its substantially more out-of-distribution GC content (24% compared with 40% for *M. pneumoniae* and approximately 38% for yeast). In contrast, *M. mycoides* has higher Jensen-Shannon divergence, indicating that the models place transcriptional mass less accurately within each window.

### 3.4 S2E models rank promoter swaps but predict a compressed expression range

**Motivation** The *Yeast Toolkit* (YTK) [16] is a frequently used modular set of building blocks for yeast synthetic biology. Its characterisation and engineering make it possible to construct gene expression cassettes by combining genetic parts, most importantly promoters, coding sequences, and terminators. By selecting promoters of different "strengths", characterised by fluorescence, the authors show that gene expression can be tuned across a very wide range, from no change above background to a thousand-fold increase in fluorescence. We investigated the models zero-shot for two reasons: (1) S2E models almost never see the same gene with different promoters during training (except for a handful of homologous genes), and (2) we previously observed that the dynamic range predicted for MPRA assays seemed suspiciously narrow.

**Task** Models are tasked with predicting the expression of two fluorescent genes (mRuby and Venus), each driven by a set of 19 promoters obtained from the native yeast genome (e.g., pTDH3).

**Data** We obtain plate-reader fluorescence measurements for 19 promoters and two fluorescent genes (the median across four replicates per promoter-gene pair) from *Lee et al.* [16]. Each data point is fluorescence normalised by biomass (OD600) and recorded as fold change over background fluorescence.

**Metrics** We reconstruct sequence cassettes comprising a promoter and coding sequence to predict RNA-seq coverage. The coding-sequence bins are then summed to obtain a single gene expression value.

$$X = \{\text{concat}(\text{upstream\_DNA}, p, r, \text{downstream\_DNA}) \mid p \in \mathcal{P}, r \in \{\text{mRuby2}, \text{Venus}\}\} \quad (9)$$

$$\hat{Y} = \left\{ \sum_{i \in I} f(x)_i \text{ where } I = [\text{start} : \text{stop}] \mid x \in X \right\} \quad (10)$$

where $x$ is the reconstructed sequence for an experimentally measured (promoter $p$, reporter coding sequence $r$) pair in the set $M$, with fixed upstream and downstream flanking DNA sequences. $f(x)_i$ is the predicted coverage value at base-pair position $i$, where $I$ is the set of all positions from the start to the end of the coding sequence.

The predicted gene expression value $\hat{y}$ is then compared with experimental fluorescence fold-change-over-background values (Pearson $r$ and Spearman $\rho$). We further compare the observed and predicted min–max ranges to determine the extent to which predictions follow the observed dynamic range. For promoter $p$ and reporter $r$, let $L_{p,r}$ denote the median fluorescence expressed as fold change over

Table 4: Results on the *Lee et al.* [16] YTK promoter benchmark, reported separately for each reporter CDS ($n = 19$ promoters per reporter). The observed range is the max/min ratio of median fold-over-background fluorescence; the predicted range is the max/min ratio of raw reporter-CDS coverage sums. Pearson $r$ is calculated after $\log_{10}$ transformation; $\rho$ denotes Spearman rank correlation. Bold denotes the better model based on unrounded values.

| Metric | **mRuby2** | | **Venus** | |
|---|---|---|---|---|
| | Shorkie | Yorzoi | Shorkie | Yorzoi |
| Observed range[2] | 66× | 66× | 481× | 481× |
| Predicted range | 2.6× | 1.9× | 2.2× | 2.2× |
| Pearson $r$ | **0.50** | 0.41 | **0.78** | 0.49 |
| Spearman $\rho$ | **0.52** | 0.47 | **0.76** | 0.46 |

the non-fluorescent-cell background. For each reporter, the observed and predicted max/min range ratios are

$$F_{\text{obs},r} = \frac{\max_{p\in\mathcal{P}} L_{p,r}}{\min_{p\in\mathcal{P}} L_{p,r}}, \quad F_{\text{pred},r} = \frac{\max_{p\in\mathcal{P}} S_{p,r}}{\min_{p\in\mathcal{P}} S_{p,r}}. \tag{11}$$

The observed and predicted extrema are selected independently. Thus, the promoter producing the predicted maximum or minimum need not be the promoter producing the corresponding experimental extremum. The range ratios are converted into base-10 logarithmic spans:

$$D_{\text{obs},r} = \log_{10} F_{\text{obs},r}, \quad D_{\text{pred},r} = \log_{10} F_{\text{pred},r}. \tag{12}$$

**Results** As with other gene-proximal genetic perturbations, both models approximately recover the ranking of promoter strengths, and Shorkie outperforms Yorzoi for both reporter genes (see Table 4). However, their predicted dynamic expression ranges are many times smaller than the observed gene expression ranges. The observed fluorescence ranges are 66-fold and 481-fold for mRuby and Venus, respectively, while the predicted transcript-abundance range is around two-fold for both models and both reporter genes. *Lee et al.* did not collect the experimental data necessary to determine the true dynamic range of transcript abundance. Thus, an unconfounded comparison between the dynamic ranges is not possible because switching promoters also changes the 5′ untranslated region, which is known to modify translation efficiency and could therefore modulate the observed fluorescence range. However, the difference between the dynamic ranges is so large that it seems implausible that differences in protein abundance caused by 5′ UTR changes alone would explain the gap. An alternative explanation is that the models indeed predict a much narrower range of gene expression. This may be partly because the models never see the same gene with different promoters during training.

### 3.5 S2E models weakly distinguish eQTLs from matched variants

**Motivation** A particularly difficult test of how sequence regulates gene expression is predicting the effect of a single-nucleotide change in the input sequence on the expression of a gene. This task is challenging because none of the evaluated models sees sequences with only one or a few nucleotides changed during training.

**Task** We use a collection of expression quantitative loci (eQTLs) and ask our models to classify single-nucleotide variants into two classes: eQTL or non-eQTL. Here, an eQTL is a (gene, variant) pair in which the presence of the variant is associated with a significant change in the expression of the selected gene. This benchmark task was adapted from [6].

**Data** We derive eQTLs from two studies, *Caudal et al.* [17] and *Kita et al.* [18], which use two yeast collections. *Caudal et al.* test gene–variant associations in the 1,011-yeast-genome collection, while *Kita et al.* use a smaller set of 86 yeast isolates. Variant–gene pairs are further filtered so that the variant and associated gene fit within the coverage windows of both models. This yields 1,850 and 605 SNP–eQTLs for [17] and [18], respectively. As described in [6], we build four sets of distance-

[2] The observed range is derived from experimental fluorescence measurements and is model-independent.

Table 5: Results on eQTL benchmark tasks. Values are means across four negative sets; standard errors (in parentheses) are in units of $10^{-3}$.

| Model | Kita (AUROC) | Kita (AUPRC) | Caudal (AUROC) | Caudal (AUPRC) |
|---|---|---|---|---|
| Shorkie | **0.67 (8.7)** | **0.64 (9.7)** | **0.57 (1.5)** | **0.58 (2.2)** |
| Yorzoi | 0.60 (2.8) | 0.57 (4.8) | 0.53 (0.7) | 0.54 (1.5) |

matched negative *non-eQTL* variant–gene pairs by matching each eQTL pair with a distance-matched variant from the same studies that is not an eQTL.

**Metrics** For each (variant $v$, gene $g$) pair (a true eQTL or a matched negative), we make two predictions: one for the reference sequence ($x_{\text{ref}}$) and one for the sequence containing variant $v$ ($x_{\text{alt}}$). Under the assumption that eQTLs should produce larger changes in gene expression than non-QTL variants, we compute the log-fold change for gene $g$ between $x_{\text{ref}}$ and $x_{\text{alt}}$:

$$s(v, g) = |\log_2(f_{i:j}(x_{\text{alt}}) + 1) - \log_2(f_{i:j}(x_{\text{ref}}) + 1)| \tag{13}$$

where $i$ and $j$ are the start and end positions of gene $g$'s coding sequence in the model output. We use $s(v, g)$ to discriminate between the classes (with $s(v, g) > \text{threshold}$ classified as eQTL). We then compute AUROC and AUPRC independently for the four negative sets and report the mean AUROC/AUPRC and standard error.

**Results** Shorkie and Yorzoi both show slightly better-than-random performance at distinguishing eQTL (variant, gene) pairs from non-eQTL pairs (see Table 5). On *Kita et al.* in particular, Shorkie shows some ability to distinguish non-eQTLs from eQTLs. Additional statistical fine-mapping might help move beyond effect direction and determine the models' ability to predict effect sizes as well.

## 4 Discussion

Across eleven benchmark tasks, Shorkie and Yorzoi generalise best to short, gene-proximal variants, particularly in promoters and terminators. Both models recover weaker but meaningful signals for synonymous variants and eQTLs. However, accurate ranking does not ensure quantitative calibration. In the YTK promoter assay, both models approximately recover promoter ordering while predicting only a two-fold expression range, compared with observed fluorescence ranges of 66-fold and 481-fold. Although fluorescence is also affected by translation, this discrepancy suggests that the models substantially underestimate effect magnitude.

The largest failures occur when fixed expression cassettes or native genes are moved into new genomic neighbourhoods. Performance is near random in both position-effect assays and remains weak for structural rearrangements. Large receptive fields therefore make contextual sequence available without ensuring that models learn its causal contribution. This may reflect the observational training data: most native genes occur in only one neighbourhood, allowing models to rely on correlated proximal features instead.

Neither model is consistently superior. Shorkie generally performs better on promoter variants and eQTLs, whereas Yorzoi leads on structural rearrangements, terminators, and foreign bacterial chromosomes. Because the models differ in architecture, training data, receptive field, and output assays, these differences require controlled ablations before they can be attributed to specific design choices.

Overall, current models are useful for prioritising local variants in matched contexts but remain unreliable for predicting absolute expression or the consequences of structural and contextual changes.

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# A Appendix

## A.1 Additional result: S2E models rank MPRA variants best in promoters and terminators, worst in 5′ UTRs

**Motivation** Massively parallel reporter assays (MPRAs) screen thousands to millions of DNA sequences at one specific position with a fixed reporter protein. Models that have learned the appropriate regulatory logic of gene expression in those regions should be able to rank the different sequences correctly.

**Task** Here, we examine four different datasets, each probing a different part of regulatory logic and benchmarked in the same way. We predict expression for a sequence construct $i$ and compute gene expression as the sum of the coding-sequence bins. This is then correlated with the true fluorescence measurements. The datasets investigate promoter, 5′ untranslated region, coding sequence, and terminator variants.

**Data** We use four different datasets from [11], [19], [20], and [21]. *Rafi et al.* [11] provide a dataset of approximately 70,000 80-bp promoter sequences, each embedded in a plasmid-based promoter scaffold that drives the transcription of a fluorescent protein. Cells are sorted into 18 bins by fluorescence-activated cell sorting (FACS). The cell-count-weighted mean bin is z-normalised and then reported as a proxy for *reporter expression*. *Cuperus et al.* [19] engineer 500,000 5′-untranslated-region (5′ UTR) variants and measure gene expression in a pooled growth assay. We note that variants in the 5′ UTR also alter translation efficiency, which no model currently captures. However, variants can, for example, alter mRNA stability, which is within the theoretical capabilities of current models. *Chen et al.* [20] perturb small stretches (up to 20 bp) of the coding sequences of two genes, GFP and TDH3, while keeping their amino acid sequences unchanged. Their proxy for gene expression is the ratio of RNA-seq reads to DNA-seq reads for each variant in the pool. *Shalem et al.* [21] screen a collection of thousands of terminator variants in a similar manner to [11]. Terminator variants are known to alter transcription stop sites and transcript isoform distributions, resulting, for example, in different mRNA stabilities.

**Metrics** Suppose $V$ is a set of variable sequences for a given dataset. We compute predictions $\hat{Y}$ for a set of constructs $C$. Each construct is a concatenation of the *fixed upstream sequence*, *variable sequence v*, and *fixed downstream sequence*. Depending on the dataset, the reporter protein is located in either the upstream or downstream sequence, or both.

$$C = \{\text{concat}(\text{upstream\_fixed}, v, \text{downstream\_fixed}) \mid v \in V\} \tag{14}$$

$$\hat{Y} = \left\{ f_{i:j}(c) \mid c \in C \right\} \tag{15}$$

where $i$ and $j$ are the start and end positions of the reporter gene's coding sequence. Given the set $Y$ of experimentally measured reporter-expression values, we report Spearman $\rho$ and Pearson $r$ for $\left\{ (y_c, \hat{y}_c) \mid c \in C \right\}$. Both Yorzoi and Shorkie are evaluated zero-shot, i.e., they have not been trained on the respective assay data.

**Results** Broadly speaking, we observe the strongest correlations with experimentally measured gene expression for the promoter and terminator assays ($r$ and $\rho$ around 0.7 ± 0.07), slightly weaker correlations for synonymous codon sequences, and the weakest for the 5′ UTR MPRA. *Shorkie* outperforms *Yorzoi* on the promoter MPRA. We speculate that this is mostly attributable to its masked-language-modelling pretraining procedure, since promoter motifs are well conserved across yeast species. To contextualise the result, we also report the performance of a supervised baseline, *DREAM-RNN* [11], evaluated on the same test set but trained on a much larger set of sequences from the same experiment. This baseline demonstrates a substantial margin over both zero-shot approaches ($\Delta r$ =0.21 compared with Shorkie).

In contrast, terminator motifs are more cryptic and less conserved. We speculate that *Yorzoi's* advantage over *Shorkie* on the terminator MPRA stems from its training on genomically rearranged sequences, in which some coding sequences are seen multiple times with different terminator sequences. When evaluating synonymous coding sequences, we chose to add two zero-shot models that use only the coding sequence to predict gene expression: *CodonTransformer* [22] and the *Codon Adaptation Index* (CAI), with the latter being a simple non-parametric formula. Surprisingly, CAI

Table 6: Results on MPRA benchmark tasks. Bold denotes the best zero-shot model for [11] and the best-performing evaluated method for Shalem, Chen, and Cuperus. Boldface is determined from unrounded values. DREAM-RNN is a supervised in-distribution reference. $r$ refers to Pearson correlation and $\rho$ to Spearman rank correlation. *CAI* is short for *Codon Adaptation Index*, and CT is short for *Codon Transformer*. *DREAM-RNN* is a supervised baseline and serves only as a reference point; it is not directly comparable.

| Benchmark | Metric | CAI | CT | Shorkie | Yorzoi | DREAM-RNN |
|---|---|---|---|---|---|---|
| **Rafi / deBoer promoter MPRA** | | | | | | |
| Overall | Pearson | — | — | **0.76** | 0.61 | 0.97 |
| Overall | Spearman | — | — | **0.77** | 0.63 | 0.98 |
| **Shalem terminator MPRA** | | | | | | |
| Overall | Pearson | — | — | 0.64 | **0.71** | — |
| Overall | Spearman | — | — | 0.65 | **0.71** | — |
| **Chen synonymous MPRA** | | | | | | |
| GFP R1 | Pearson | 0.29 | 0.18 | **0.30** | 0.28 | — |
| GFP R1 | Spearman | 0.32 | 0.29 | **0.39** | 0.36 | — |
| GFP R2 | Pearson | 0.28 | **0.57** | 0.57 | 0.56 | — |
| GFP R2 | Spearman | 0.28 | **0.59** | 0.59 | 0.57 | — |
| TDH3 | Pearson | **0.68** | 0.36 | 0.53 | 0.61 | — |
| TDH3 | Spearman | **0.39** | 0.19 | 0.30 | 0.23 | — |
| **Cuperus 5′-UTR MPRA** | | | | | | |
| Overall | Spearman | — | — | **0.25** | 0.12 | — |

and CodonTransformer outperform Shorkie and Yorzoi on two of the three subtasks (GFP region 2 and TDH3), while Yorzoi and Shorkie have the best average performance.

The *Cuperus et al.* [19] assay is the least predictable. We suspect that even perfect RNA-seq coverage prediction would not yield a very high correlation in this assay because the 5′ UTR sequence, especially around the start codon, has a major influence on translation efficiency, which current models do not capture directly. A second reason lies in the assay itself: a pooled growth experiment in HIS3-knockout strains, for which it is unclear whether enrichment in the pool remains correlated with protein expression above a certain protein abundance. Histidine biosynthesis is known to be product-inhibited; i.e., overexpression of HIS3 increases histidine synthesis only up to a certain point. Future models that predict protein abundance in addition to transcript abundance might improve upon the results shown here.

### A.2 Model training, data and architecture details (*Shorkie* and *Yorzoi*)

The two sequence-to-expression models that we investigate in the benchmark are *Shorkie* (*Chao et al.* [6]) and *Yorzoi* (*Schneider et al.* [7]). Both are trained on sequences from the model organism *S. cerevisiae* (baker's yeast). Although both models predict RNA-seq tracks from yeast DNA sequences, there are a few differences that make them worth comparing. First, Shorkie fine-tunes ShorkieLM, a masked language model trained on 165 yeast species, to predict expression and ChIP-seq tracks, and has an input sequence window of approximately 16 kilobase pairs (kbp). Its predicted RNA-seq tracks are unstranded. Yorzoi, on the other hand, is trained on long-read RNA-sequencing data; its training data supplement native yeast DNA sequences with human DNA profiled in yeast and structurally rearranged yeast sequences. It has a shorter input sequence window of approximately 5 kbp. Its predicted RNA-seq tracks are stranded, i.e., one track for the forward strand and one for the reverse strand in each RNA-seq experiment. For Yorzoi, a pair of tracks represents the *coverage* of the two DNA strands, whereas Shorkie predicts only unstranded RNA-seq tracks, and thus one track represents one experiment. Both models apply additional symmetric cropping to the output such that $Y \in \mathbb{R}^{T,K}$, where $K < L$. Thus, they predict track values only for the central $K$ positions in the input sequence.

### A.3 MPRA marginalisation

**TODO**: I think this is not explained at all

### A.4 Brooks LFC calculation

**TODO**:

- pred calculation is glossed over.
- maybe we need a general description

### A.5 *Hong et al.* boundary distance definition

**TODO**: short explainer and citation of the Micro-C paper where they actually define this. *Boundary distance* is defined as $\text{boundary\_distance}_i = \min_j |\text{insertion\_site}_i - \text{boundary}_j|$, where $\text{boundary}_j \in B$ and the set of boundaries $B$ is

$$B = \underset{i}{\text{argmin}}\{C(i) \mid i \in [L]\} \tag{16}$$

$$C(i) = \sum_{(x,y)} \mathbb{1}[x < i < y]\mathbb{1}[500 \leq y - x \leq 10,000] \tag{17}$$

where $i$ is any position in a genome of length $L$.

### A.6 [More detailed Explanation of the eQTL origination and filtering]

## B Use of LLMs for creating this work

LLMs were involved in creating this work through coding agents that helped to write, debug, and test the code produced for this benchmark. LLMs were also used to write parts of the code documentation (`docs/` in the repository). Additionally, they were used to improve the fluency and correct the grammar and spelling of the manuscript.